Team A - Ames Housing Project Plan

Project Overview

This project focuses on analyzing housing market data and predicting residential property valuations using the Ames Housing Dataset. The predictive modeling direction is guided by feedback from Doris to benchmark linear and ensemble models.

- **Dataset Scope:** Ames Housing Dataset containing 1,460 records and ~79 explanatory variables detailing residential home sales.
- **Core Objective:** Predict `SalePrice` through rigorous comparative modeling.
- **ML Strategy (Doris's Feedback):** Benchmark baseline performance using Linear Regression versus an ensemble approach via Random Forest Regressor to capture non-linear feature interactions and optimize predictive accuracy.

Methodology

The pipeline follows an end-to-end analytics workflow combining relational querying with exploratory and predictive data science libraries.

- **Data Extraction & Ingestion:** Utilize SQL for querying raw tables, filtering subsets, and executing preliminary table joins and schema validation.
- **Exploratory Data Analysis (EDA):** Execute deep-dive feature analysis in Python (utilizing pandas, NumPy, seaborn, and matplotlib) to detect outliers, assess multi-collinearity, and guide transformation steps.
- **Feature Engineering & Preprocessing:** Address missing values, apply one-hot and ordinal encoding, and normalize skewed distributions for both linear and tree-based consumption.
- **Model Development & Evaluation:** Train, tune, and cross-validate Linear Regression and Random Forest models using scikit-learn, evaluating performance via RMSE and R^2 metrics.
- **Business Intelligence & Reporting:** Design an interactive Power BI dashboard for executive reporting, stakeholder scenario analysis, and model performance inspection.

Beginner Execution Guide & Tech Breakdown

Welcome to the execution guide! You've built solid foundations over the past 6 weeks. Remember, our team consists of Favour, Trizah, Tanran, and Latita (our only boy).

Work together, support each other through debugging, and divide up these actionable tasks according to what you want to practice most.

SQL Tasks (Data Extraction):

- **Task 1:** Write a SQL query to inspect the schema and retrieve all records where the zoning is residential, filtering out abnormal sales or incomplete deeds (e.g. `SaleCondition = 'Normal'`).
- **Task 2:** Write a SQL query using aggregation (GROUP BY) to calculate the count of homes, average sale price, and median price per square foot across each neighborhood.
- **Task 3:** Formulate a query to identify missing data or NULL values across key property attributes (such as LotFrontage, GarageType, and Basement specs) to guide Python data cleaning.

Python Tasks (EDA - Exploratory Data Analysis):

- **Task 1:** Calculate the statistical distribution, skewness, and kurtosis of `SalePrice`, and plot a histogram with a KDE curve to check if a log transformation is required.
- **Task 2:** Find the top 5 neighborhoods with the highest median sale prices and plot a horizontal bar chart using seaborn.
- **Task 3:** Plot a scatter plot of Above Grade Living Area (GrLivArea) vs. `SalePrice` to detect extreme outliers (homes with huge square footage but low prices).
- **Task 4:** Engineer new features (e.g. TotalSqFt = TotalBsmtSF + 1stFlrSF + 2ndFlrSF; HouseAge = YrSold - YearBuilt) and handle missing categorical data by filling NA with 'None'.

Machine Learning Tasks (Prediction):

- **Task 1:** Break down the `SalePrice` prediction using Linear Regression as your baseline model, splitting data into an 80/20 train-test split, and compute baseline RMSE and R^2 metrics.
- **Task 2:** Compare the results using Random Forest Regressor and tune basic hyperparameters (e.g. n_estimators=100, max_depth=10) to see if ensemble decision trees capture non-linear patterns better.
- **Task 3:** Explain the evaluation metrics (MAE, RMSE, and R^2) side-by-side in a summary table and extract the top 10 most influential features using feature_importances_.

Power BI Tasks (Dashboarding):

- **Task 1:** Import cleansed dataset and model predictions; create simple DAX measures for Average Sale Price, Total Houses Sold, and Prediction Error (Actual - Predicted).
- **Task 2:** Put a Clustered Bar Chart on Page 1: set the Y-axis to Neighborhood, X-axis to Average Sale Price, and add a Slicer for Overall Quality rating (1 to 10).
- **Task 3:** Put a Scatter Chart on Page 2: set the X-axis to Actual Sale Price, Y-axis to Predicted Sale Price, and add a 45-degree reference line to visually assess prediction accuracy.